Annexure-B

BCA

Fifth and Sixth Semesters Syllabus, 2026-27

University of Mysore

Semester	Course No.	Theory/ Practical	Credits	L-T-P	No. of Hours	Paper Title	Marks	
							SEE	CIE
VI	CAM61T	**Theory**	03	3-0-0	03	**Scientific Computing Techniques**	80	20
	CAM61P	**Practical**	02	0-0-2	04	**Scientific Computing Techniques Lab**	40	10
	CAM62T	**Theory**	03	3-0-0	03	**Big Data Analytics**	80	20
	CAM62P	**Practical**	02	0-0-2	04	**Big Data Analytics Lab**	40	10
	CAM63T	**Theory**	03	3-0-0	03	**Web Content Management System**	80	20
	CAM63P	**Practical**	02	0-0-2	04	**Web Based Mini Project**	40	10
	CAM64T	**Theory**	03	3-0-0	03	**Software Project Management**	80	20
	CAM65T	**Theory**	03	3-0-0	03	**Mobile Communication**	80	20
	CACP61P	**Practical (Compulsory Paper)**	02	0-0-2	04	**Power BI**	40	10

Semester: VI

Course Code:	**Course Title: Scientific Computing Techniques**
Course Credits: 03 (3:0:0)	**Hours/Week:** 03
Total Contact Hours: 44	**Formative Assessment Marks:** 20
Exam Marks: 80	**Exam Duration:** 03

Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

CLO 1: Understand the concepts of computer arithmetic including fixed-point and floatingpoint representation, normalization, and error analysis in numerical computations.
CLO 2: Apply iterative methods such as Bisection, Newton–Raphson, Regula Falsi, and Secant methods to find roots of polynomial equations.
CLO 3: Solve ordinary differential equations and numerical integration problems using methods such as Euler's method, Runge–Kutta methods, Simpson's rule, and the Trapezoidal rule.
CLO 4: Analyze and compute solutions of simultaneous linear equations using numerical techniques such as Gauss Elimination, Gauss–Jordan, LU Decomposition, and Gauss–Seidel methods.

Course Pedagogy

1. Conceptual Lectures and Problem-solving Sessions to explain numerical methods, floating-point arithmetic, and error analysis.
2. Worked Examples and Tutorial Exercises to practice iterative methods, numerical integration, and differential equation solutions.
3. Assignments and Analytical Comparisons where students compare direct and iterative numerical methods for efficiency and accuracy.

Content	**Hours**
Unit 1:	
Computer Arithmetic**:** Need of Computer Oriented Numerical Analysis, Fixed and Floating-point representation of numbers, Normalization, arithmetic operations with normalization, consequences of normalized floating-point representation of numbers, Concept of Zero in floating point, Errors in computing, Measure of errors	11
Unit 2:	

Finding the roots of an equation**:** Iterative method: Introduction, types of equation Methods of solution, Beginning an iterative method, Bisection method, Newton Raphson method,	11
Regula Falsi method, Secant method. Comparison of Iterative methods. (Problems only on polynomial equations)	
Unit 3:	
Ordinary differential equations: Euler's method, Taylor series method, Runge Kutta II and IV order methods. Numerical Integration: Introduction, Simpson's 1/3 and 3/8 rule, Trapezoidal rule.	11
Unit 4:	
Solving simultaneous linear equations: Introduction, Gauss Elimination method, pivoting, ill-conditioned equations, Gauss Jordon method, LU Decomposition method and GaussSeidel iterative method. Comparison of direct and iterative methods.	11

Learning Resources:

1. Computer Oriented Numerical Methods by Rajaraman. V.
2. Fundamentals of Mathematical Statistics by Gupta and Kapoor (Sultan Chand).
3. Probability and Statistics for engineers and scientists by Ronald E. Walpole and Raymond H Mayers.
4. Mathematical Statistics by John Freund (Prentice Hall India Pvt. Ltd.)
5. Numerical Methods by Jain M.K., S.R.K. Iyengar and R.K. Jain

Semester: VI

Course Code:	**Course Title: Scientific Computing Techniques Lab**
Course Credits: 02 (0:0:2)	**Hours/Week:** 04
Total Contact Hours: 60	**Formative Assessment Marks:** 10
Exam Marks: 40	**Exam Duration:** 03

Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

CLO 1: Implement numerical methods such as Bisection, Regula Falsi, and Newton–Raphson to determine the roots of algebraic equations using programming.
CLO 2: Apply numerical techniques like Euler's method and Runge–Kutta methods to solve ordinary differential equations computationally.
CLO 3: Compute numerical integration using methods such as Trapezoidal rule, Simpson's 1/3 rule, and Simpson's 3/8 rule.
CLO 4: Develop programs to solve systems of simultaneous linear equations using methods like Gauss Elimination, Gauss–Jordan, and Gauss–Seidel.

Course Pedagogy

1. Demonstration-based Laboratory Sessions where instructors explain numerical algorithms and their implementation in C++ programming.
2. Hands-on Programming Exercises enabling students to write, compile, and test programs for various numerical methods.
3. Problem-solving and Debugging Activities to help students understand computational errors, accuracy, and convergence of numerical methods.
4. Lab Assignments and Mini Experiments where students modify programs, test with different inputs, and compare results of different numerical techniques.

List of Practical Experiments:

PART-A

1. Program to solve the given equation x*x-14=0 by using Bisection method.
2. Program to solve the given equation x*x+5*x-6=0 by using Regula Falsi method.
3. Program to solve the given equation x*x*x*-12=0 by using Newton Raphson method.
4. Program to solve the given equation dy/dx=1+y*y where y(0)=1, h=0.1, find y(0.4) using Eulers method.
5. Program to solve the given equation dy/dx=x*y where y(1)=2, h=0.3, find y(1.4) by using

Runge-kutta’s II order method.

6. Program to solve the given equation dy/dx= x*y where y(1)=2, h=0.3, find y(1.4) by using Runge-kutta’s IV order method.

PART-B

7. Program to solve the given equation $\int dx/(1+x)$ where, a=0, b=1, n=10 by using Trapezoidal method.
8. Program to solve the given equation $\int \sin x \, dx$ where a=0, b=π /2, n=6 by using Simpson’s 1/3 rule.
9. Program to solve the given equation $\int \sin x \, dx$ where a=0, b=π /2, n=6 by using Simpson’s 3/8 rule.
10. Program to solve the following set of simultaneous equation x+y+4z=12, 8x3y+2z=20,4x+11y-z=33 using the Guass Elimination method.
11. Program to solve the following set of simultaneous equation $2x_1+6x_2-x_3=-14$,$5x_1-x_2+2x_3=29$,$-3x_1-4x_2+x_3=4$ using Guass Jordon method.
12. Program to solve the following set of simultaneous equation $2x_1-x_2+x_3=5$, $x_1+3x_22x_3=7$, $x_1+2x_2+3x_3=10$ using Guass Seidel method.

Evaluation Scheme for Lab Examination:

Assessment Criteria		**Marks**
Writing	One Program from Part A	15
	One Program from Part B	15
Execution	Any one of the Written Program	5
Viva Voce		5
Total		**40**

Semester: VI

Course code:	**Course title: Big Data Analytics**
Course credits:03 (3:0:0)	**Hours/Week: 3**
Total contact class: 44	**Formative assessment marks :20**
Exam marks: 80	**Exam duration: 3 Hours**

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

CLO 1: Identify and list various Big Data concepts, tools and applications.
CLO 2: Describe Hadoop architecture and ecosystem.
CLO 3: Analyze MapReduce data processing.
CLO 4: Understand NoSQL databases and gain the basic knowledge of Hive queries. **CLO 5.** Understand social network analysis basics.

Course Pedagogy

1. Interactive Lecture Method in Classroom teaching using board and PPT to explain Big Data concepts and Hadoop framework.
2. Demonstration of Hadoop ecosystem, Hive queries and MapReduce concepts.
3. Interactive Learning through Group discussions, quizzes and seminars on Big Data applications.
4. Assignment/Case Study Method for Problem solving tasks and case studies on real-world Big Data applications.

Course contents	**Hours**
Unit 1:	
Introduction to Big Data, Characteristics of Big Data (5 V's), Types of Big Data, Traditional Data vs Big Data, Evolution of Big Data, Challenges in Big Data Analytics: Big data Analytics-definition, Classification of Analytics, Importance of Big Data Analytics, Technologies used in Big data Environments, Few Top Analytical Tools, NoSQL, Hadoop	11

Unit 2:	
Introduction to Hadoop, Features – Advantages – Versions – Overview of Hadoop, Hadoop Components, Hadoop Limitations, Hadoop vs. SQL, HDFS (Hadoop Distributed File System), Processing data with Hadoop, YARN Resource Management. Introduction to MapReduce, advantages of MapReduce, MapReduce Framework, MapReduce Architecture, Working of MapReduce, Phases of MapReduce- Map phase, Shuffle phase, Reduce phase	11
UNIT-3:	
Introduction to NoSQL, Need for NoSQL databases, Types of NoSQL databases, Key-valu stores, Document databases, Column family databases, Graph databases, Comparison of SQL vs NoSQL. Introduction to MongoDB, Basic MongoDB operations, Data ingestion concepts, Data processing techniques. Introduction to Apache Spark	11
UNIT-4:	
Introduction to Hive, Hive, Architecture, Hive Data Types, Hive Query Language (HQL). Introduction to Social Network Mining, Applications, Social Networks as Graphs, Types of Social Networks, Clustering of Social Graphs, Community Discovery, Recommender Systems	11

Learning Resources:

1. Seema Acharya and Subhashini Chellappan "Big data and Analytics" Wiley India Publishers, 2nd Edition, 2019.
2. Rajkamal and Preeti Saxena, "Big Data Analytics, Introduction to Hadoop, Spark and Machine Learning", McGraw Hill Publication, 2019.
3. Understanding Big Data, McGraw-Hill.
4. Adam Shook and Donald Mine, "MapReduce Design Patterns: Building Effective Algorithms and Analytics for Hadoop and Other Systems" - O'Reilly 2012
5. Tom White, "Hadoop: The Definitive Guide" 4th Edition, O'reilly Media, 2015.
6. Thomas Erl, Wajid Khattak, and Paul Buhler, Big Data Fundamentals: Concepts, Drivers & Techniques, Pearson India Education Service Pvt. Ltd., 1st Edition, 2016

Semester: VI

Course code:	Course title: Big Data Analytics Lab
Course credits:02 (0:0:2)	**Hours/Week: 4**
Total contact class: 60	**Formative assessment marks :10**
Exam marks: 40	**Exam duration: 3 Hours**

Course Learning Objectives (CLOs)

Upon successful completion of this course, students will be able to:

CLO 1: Understand Big Data environment and configure basic Hadoop ecosystem tools.
CLO 2: Develop basic MapReduce programs for data processing tasks such as word count, maximum, minimum and data analysis.
CLO 3: Perform NoSQL database operations using MongoDB including CRUD operations and data retrieval.
CLO 4: Apply Hive queries for managing and analyzing large datasets.
CLO 5: Implement simple Big Data applications using MongoDB and Hadoop tools.

Course Pedagogy

1. Demonstrates Hadoop, MongoDB and Hive commands.
2. Hands-on Practical sessions to execute programs individually in the lab.
3. Problem Solving Approach **to** solve real data processing problems using MapReduce.
4. Mini Assignments on MongoDB queries and Hive operations.

List of Experiments:

Part A

1. Install Hadoop (Standalone/Pseudo Mode) or study Hadoop installation procedure and write the steps to start Hadoop services.
2. Write a MapReduce program to count words in a given text file (Word Count Problem).
3. Write a MapReduce program to find the maximum number from a given dataset.
4. Write a MapReduce program to find the minimum number from a given dataset.
5. Write a MapReduce program to find total and average marks of students from a given dataset.
6. Write a MapReduce program to count frequency of numbers or names in a dataset.
7. Write a MapReduce program to find maximum temperature from a weather dataset.

Part B

8. Create a database and collection in MongoDB and display the created database.
9. Insert student records into MongoDB collection and perform query operations: i. Display all records
   ii. Display records with condition iii.Display
   selected fields
10. Implement update and delete operations in MongoDB.
11. Implement NoSQL database operations using MongoDB:
    i. CRUD operations (Create, Read, Update, Delete)
    ii. Working with Arrays in MongoDB
12. Implement MongoDB functions:

i. Count
ii. Sort
iii. Limit
iv. Skip
v. Aggregate functions

13. Create Hive table and load data. Perform basic Hive queries:
    i. Select records
    ii. Where condition
    iii. Order by clause
14. Implement a simple application that stores big data in MongoDB
    (Example: Student database, Employee database, Product database).

Evaluation Scheme for Lab Examination:

Assessment Criteria		Marks
Writing	One Program from Part A	15
	One Program from Part B	15
Execution	Any one of the Written Program	5
Viva Voce		5
Total		**40**

Semester: VI

Course Code:	**Course Title: Web Content Management System**

Course Credits: 03 (3:0:0)	**Hours/Week:** 03
Total Contact Hours: 44	**Formative Assessment Marks:** 20
Exam Marks: 80	**Exam Duration:** 03

Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

CLO 1: Explain the principles of web content development, content formats, and guidelines for creating effective digital multimedia content.
CLO 2: Develop digital media elements such as graphics, audio, presentations, and multilingual web content using appropriate tools.
CLO 3: Design and manage dynamic web content including blogs, wiki sites, and websites using CSS and content management techniques.
CLO 4: Apply modern web technologies and platforms such as Moodle, Joomla, and Drupal for course creation, content management, and online publishing.

Course Pedagogy

1. Interactive Lectures and Demonstrations to introduce web content development concepts, multimedia tools, and CMS platforms.
2. Case study-based Learning where students develop multimedia web content or create a small website using CMS platforms.
3. Student Presentations and Collaborative Activities using presentation tools, screen casting techniques, and online learning platforms.

Content	Hours
Unit 1:	
Web Content Development and Management, Content Types and Formats, Norms and Guidelines of Content Development, Creating Digital Graphics, Audio Production and Editing.	11
Unit 2:	
Web Hosting and Managing Multimedia Content, Creating and Maintaining a Wiki Site. Presentation Software Part I, Presentation Software Part II, Screen casting Tools and Techniques, Multilingual Content Development.	11
Unit 3:	
Planning and Developing Dynamic Web Content Sites, Website Design Using CSS Creating and Maintaining a WIKI Site, Creating and Managing a Blog Site, E- Publication Concept, E- Pub Tools, Simulation and Virtual Reality Applications.	11
Unit 4:	

Creating 2D and 3 D Animations. Introduction to Moodle, Creating a New Course and Uploading. Create and Add Assessment, Add and Enroll User and Discussion Forum,	11
Content Management System: Joomla, Content Management System: Drupal	

Learning Resources:

1. Web Content Management: Systems, Features, and Best Practices 1st Edition by Deane Barker.
2. Content Management Bible (2nd Edition) 2nd Edition by Bob Boiko.
3. Moodle for Learning Management System (LMS): A Practical and Visual Guidebook of Administrator and Instructor for Distance Education Paperback – October 12, 2020 by James Koo
4. Using Joomla!: Efficiently Build and Manage Custom Websites 2nd Edition by Ron Severdia

Semester: VI

Course Code:	Course Title: Web Based Mini Project
Course Credits: 02 (0:0:2)	**Hours/Week:** 04
Total Contact Hours: 60	**Formative Assessment Marks:** 10
Exam Marks: 40	**Exam Duration:** 03

Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

CLO 1: Identify and analyze real-world problems and propose suitable web-based solutions using modern web technologies.
CLO 2: Design and develop dynamic and responsive web applications using technologies such as HTML, CSS, JavaScript, and server-side scripting (e.g., PHP).
CLO 3: Apply software development life cycle (SDLC) phases including planning, design, implementation, testing, and deployment in a mini project.
CLO 4: Demonstrate the ability to integrate front-end and back-end components to build a complete functional web application.
CLO 5: Work effectively in teams, manage project tasks, and demonstrate professional ethics in project development.
CLO 6: Prepare technical documentation and present the project effectively using appropriate tools and communication skills.

Course Pedagogy

1. Project-Based Learning by developing a complete web application addressing realworld problems.
2. Hands-on practice through coding, debugging, testing, and deployment.
3. Identify the problems, analyze requirements, and implement feasible solutions.
4. Group-based mini projects to enhance teamwork and communication skills.

List of Web-Based Mini Projects:

1. Student Record Management System
2. Employee Details Management
3. Library Management (Mini Version)
4. Online Complaint Registration System
5. College Event Registration Portal
6. Online Assignment Submission System
7. Student Feedback System
8. Attendance Management System (Web-based)

9. Result Publishing System
10. Online Book Store
11. Food Ordering System (basic version)
12. Bus/Train Ticket Booking (demo system)
13. Online Rental System (rooms, bikes, etc.)
14. Personal Expense Tracker
15. To-Do List / Task Manager
16. Online Notes Sharing Platform
17. Weather App (API-based)
18. Online Quiz System
19. Online Voting System (with login authentication)
20. Course Recommendation System
21. Web-Based Health Monitoring System
22. Student Performance Analysis System
23. Employee Timesheet System
24. Online Survey & Feedback Analyzer

(Note: Choose any one mini project based on above theme /choose any similar kind of mini project titles)

Technology to use:
Frontend: HTML, CSS, JavaScript, Bootstrap etc.,
Backend: PHP / Python (Flask/Django) / Node.js etc.,
Database: MySQL / MongoDB etc.,

Project Guidelines:
1. Maximum 4 students shall be allowed to take up the mini project.
2. Each student will have to work for 4 hours per week in the lab premises
3. Each student shall submit his/her mini project synopsis to the concerned teacher ·
4. Each student has to carry out 2 project seminars compulsorily in project duration.

Each seminar will be considered for their internal assessment.

Scheme of valuation - 50 Marks

1. **IA – 10 Marks**
   - Synopsis - **05 Marks**
   - Seminar - **05 Marks**

2. **Dissertation – 40 Marks**
   - Documentation - 15 **Marks**
   - Presentation / Demonstration - 15 **Marks**

3. **Viva- 10 Marks**

Semester: VI

Course code:	Course title: Software Project Management
Course credits:03 (3:0:0)	**Hours/Week: 3**
Total contact class: 44	**Formative assessment marks: 20**
Exam marks: 80	**Exam duration: 3 Hours**

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to,

CLO 1: Understand software project basics.
CLO 2: Analyze the requirements and formulate appropriate plan.
CLO 3: Design Software effort estimation by adopting different project control measures activity planning
CLO 4: Use resource allocation and risk project management techniques appropriately.
CLO 5: Demonstrate Managing people and organizing teams uploading software quality management guidelines.

Course Pedagogy

1. Interactive teaching and guided discussions
2. Application-driven Learning through Case Examples
3. Collaborative Learning Activities and Team-based Tasks
4. Seminars, Reports, and Presentations

Course Contents	Hours
Unit 1: Introduction, importance of software project management, project definition, Software projects versus other types of projects, Contract management and technical project management, Activities covered by software project management, Plans, methods and methodologies, some ways of categorizing software projects, Stakeholders, Setting objectives, The business case, Project success and failure, management and management control. An overview of project planning: Select project, identify scope and objectives, identify project infrastructure, Analysis project Characteristics, identify project products and activities, estimate efforts for each activity, identify activity risks, allocate resources, Review/publicize plan, execute plan and lower levels of planning.	11

Unit 2: Project evaluation: Introduction, a business case, Project portfolio management, Evaluation of individual projects, Cost-benefit evaluation techniques, Risk	11
evaluation, Programme management, Managing the allocation of resources within programmes, Strategic programme management, Creating a programme, Aids to programme management, Benefits management. Selection of an appropriate project approach: Introduction, Build or buy, choosing methodologies and technologies, Choice of process models, Structure versus speed of delivery, The waterfall model, The spiral model, Software prototyping, Agile methods, managing iterative Processes, Selecting the most appropriate process model.	
Unit 3: Software effort estimation: Introduction, need of estimation. Problems with over and under-estimates, the basis for software estimating, Software efforts estimation techniques, Bottom-up estimating, the top-down approach and parametric models, Expert judgment, estimating by analogy. Activity planning: Introduction, objectives, Project schedules, Projects and activities, Sequencing and scheduling activities, Network planning models, Shortening the project duration, Identifying critical activities. Risk Management: Introduction, Categories of risk, framework for dealing with risk, risk identification, risk assessment, risk planning, risk management.	11
Unit 4: Resource allocation: Introduction, the nature of resources, identifying resources requirements, scheduling resources, creating critical paths, Counting the cost, being specific, Publishing the resources schedule, Cost schedules. Managing people and organizing teams: Introduction, understanding behavior, Motivation, team work, Decision Making, Leadership. Software quality: Introduction, importance, Product versus process quality management, Techniques to help enhance software quality, Quality plans.	11

Learning Resources:

1. Software Project Management, Bob Hughes, Mike Cotterell, 5th Edition, McGraw-Hill Education, 2009.
2. Project Management, Harvey Maylor, 3rd Edition, Pearson, 2004.
3. Project Management and Agile Systems, Kalpesh Ashar, 5th Edition, Vibrant Publishers, 2020.

Semester: VI

Course Code:	**Course Title: Mobile Communication**
Course Credits: 03 (3:0:0)	**Hours/Week:** 03
Total Contact Hours: 44	**Formative Assessment Marks:** 20
Exam Marks: 80	**Exam Duration:** 03

Course Learning Outcomes (CLOs)
Upon successful completion of this course, students will be able to:

CLO 1: Understand the basic concepts of wireless and mobile communication
CLO 2: Explain multiple access techniques used in mobile communication such as FDMA, TDMA, and CDMA.
CLO 3: Analyze radio propagation mechanisms and wireless transmission techniques used in mobile communication environments.
CLO 4: Focus on cellular network design concepts, including frequency reuse, handoff strategies, and channel allocation.

Course pedagogy

1. Classroom lectures to explain fundamental concepts of wireless and cellular communication systems.
2. Demonstration of wireless communication concepts such as signal transmission, cellular structure, and handoff mechanisms.
3. Case Studies: Study of real-world mobile communication systems and applications to understand practical implementations.
4. Group Discussions and Presentations.

Course Contents	**Hours**
Unit 1:	
Introduction to Mobile Communication: Evolution of mobile communication systems. **B**asic concepts of wireless communication, Cellular communication principles, Frequency reuse and channel assignment strategies, multiple access techniques such as FDMA, TDMA, and CDMA.	11
Unit 2:	
Wireless Transmission and Propagation: Free space propagation model, Radio wave propagation mechanisms, Path loss and shadowing, fading models, Doppler Effect in mobile communication. : Cellular Network Design, Cell structure and cell splitting Handoff and handover strategies	11

Unit 3:	
Traffic engineering in cellular systems, Channel allocation techniques, Capacity improvement techniques. Wireless Network Technologies**:** Architecture and operation of GSM, Overview of 3G, 4G, and 5G networks, Mobile switching and call processing, Wireless LAN and Bluetooth technologies.	11
Unit 4:	
Mobile Network Protocols and Security: Mobile IP and mobility management, Routing in mobile networks, Security issues in wireless networks, Authentication and encryption techniques, Mobile application services.	11

Learning Resources:

1. Mobile Communications – by Jochen Schiller, Publisher: Addison-Wesley
2. Wireless and Mobile Communication – by Upena Dalal and Manoj K. Shukla 3. Principles of Mobile Communication – by Gordon L. Stüber
4. Mobile Wireless Communications – by Mischa Schwartz.
5. Wireless Communications: Theory and Techniques – by Asrar U. H. Sheikh
6. Mobile Communications Handbook – edited by Jerry D. Gibson

Semester: VI

Course Code:	Course Title: Power BI
Course Credits: 02 (0:0:2)	**Hours/Week:** 04
Total Contact Hours: 60	**Formative Assessment Marks:** 10
Exam Marks: 40	**Exam Duration:** 03

Course Learning Outcomes (CLOs)

Upon successful completion of this course, students will be able to:

CLO 1: Import, clean, and transform datasets using tools such as Power Query Editor for effective data preparation.
CLO 2: Create meaningful data visualizations including bar charts, line charts, pie charts, and donut charts to represent business data.
CLO 3: Develop interactive reports and dashboards to analyze key performance indicators (KPIs) and business performance.
CLO 4: Apply data modeling techniques and DAX expressions to establish relationships between tables and generate calculated fields for advanced analysis.

Course Pedagogy

1. Hands-on Laboratory Sessions using Power BI for data import, transformation, visualization, and dashboard creation.
2. Demonstration-based Teaching where the instructor explains Power BI tools such as Power Query Editor, data modeling, and DAX functions.
3. Practical Assignments and Exercises requiring students to analyze datasets and build visual reports and dashboards.

List of Practical Experiments:

Part -A

1. Import a dataset from an Excel file and clean the data by removing duplicates and handling missing values.
2. Use Power Query Editor to transform a dataset, such as splitting columns, merging tables, or changing data types.
3. Create a bar chart to show the total sales by region.
4. Create a line chart to display the trend of sales over time.
5. Build a report that includes various visualizations, such as pie charts, bar charts, and tables, to analyze sales data.

Part -B

6. Create a dashboard that combines several key performance indicators (KPIs) to give an overview of business performance.
7. Create a relationship between tables manually through Power BI. 8. Use the DAX expression to add a new column in the data source.
9. Use the Donut Chart for Profit Analysis.
10. Analyze the more profitable day from the sales analysis table using Bar Chart in Power BI.

Evaluation Scheme for Lab Examination:

Assessment Criteria		**Marks**
Writing	One program from Part A	15
	One program from Part B	15
Execution	Any one of the Written programs	5
Viva Voce		5
Total		**40**

CIE, SEE and QP Pattern for Theory Courses (2 Credits):

Total Lecture hours per paper: 30, No. of Units: 3 (10 Hours Each)

Internal Assessment: C1 - 5 Marks, C2 - 5 Marks

Semester End Theory Exam: C3 - 40 Marks

Question paper pattern for Theory Courses (2 Credits):

Maximum Marks – 40

Part A: Answer any 5 questions out of 6: (5x2=10 Marks) Question No. 1 to 6

Part B: Answer the following questions: (10x3=30 Marks)

7.a) 7. b) or 7. c) 7. d)

8.a) 8. b) or 8. c) 8. d)

9.a) 9. b) or 9. c) 9. d)

CIE, SEE and QP Pattern for Theory Courses (3 Credits):

Total Lecture hours per paper: 44

No. of Units 4 (11 Hours Each)

Internal Assessment: C1 - 10 Marks, C2 - 10 Marks

Semester End Theory Exam: C3 - 80 Marks

Question paper pattern for Theory Courses (3 Credits):

Instructions: Answer Part-A and Part-B:

Part-A

Answer any 10 out of 12 Questions (3 Questions drawn from each unit).

Each question carries 2 Marks. (10 X 2 =20) Q.

No. 1 to Q. No. 12.

Part-B

Answer all the Questions. Each question carries 15 Marks. (4 X 15 =60) (Each question with internal choice and with maximum of 3 sub questions)

Programme Learning Outcomes (PLOs)

PLO 1: Computational Knowledge

Apply the knowledge of computing fundamentals, mathematics, and domain concepts to solve complex computing problems in areas such as operating systems, computer networks, algorithms, artificial intelligence, and data analytics.

PLO 2: Problem solving skills

Identify, formulate, and analyze complex computing problems using logical reasoning, algorithmic thinking, and appropriate computational techniques.

PLO 3: Modern Tool Usage

Select and apply modern tools, technologies, and platforms such as AI tools, data analytics software, block chain systems, and visualization tools for effective problem-solving.

PLO 4: Experimentation & Data Analysis

Conduct experiments, analyze data, and interpret results using statistical, computational, and visualization techniques to support decision-making.

PLO 5: Professional Ethics & Security

Apply ethical principles and commit to professional responsibilities, including data security, privacy, and legal compliance in computing practices.

PLO 6: Team Work

Function effectively as a member or leader in multidisciplinary teams to accomplish shared goals.

PLO 7: Project Management

Apply project management principles, including planning, scheduling, risk management, and resource allocation in software and IT projects.

Sustainable Development Goals (SDGs)

SDG 4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

SDG 5: Achieve gender equality and empower all women and girls.

SDG 9: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.

SDG 13: Take urgent action to combat climate change and its impacts.

Programme Learning Outcomes (PLOs) and Sustainable Development Goals (SDGs) - Course Compliance Matrix

Sl. No.	Course Title	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7
1	Network Security	✓	✓	✓	✓	✓	-	-
2	Network Security Lab	✓	✓	✓	✓	✓	✓	-
3	Design and Analysis of Algorithms	✓	✓	✓	✓	-	✓	-
4	Design and Analysis of Algorithms Lab	✓	✓	✓	✓	-	✓	-
5	Computer Graphics & Multimedia	✓	✓	✓	✓	-	✓	✓
6	Computer Graphics & Multimedia Lab	✓	✓	✓	✓	-	✓	✓
7	Blockchain Technology	✓	✓	✓	✓	✓	-	✓
8	Enterprise Resource Planning	✓	✓	✓	✓	✓	✓	✓
9	Fundamentals of Artificial Intelligence	✓	✓	✓	✓	✓	-	✓
10	AI Tools and its applications	✓	✓	✓	✓	✓	✓	✓
11	Scientific Computing Techniques	✓	✓	-	✓	-	-	-
12	Scientific Computing Techniques Lab	✓	✓	✓	✓	-	✓	-
13	Big Data Analytics	✓	✓	✓	✓	-	-	✓
14	Big Data Analytics Lab	✓	✓	✓	✓	-	✓	✓
15	Web Content Management System	✓	✓	✓	✓	-	✓	✓
16	Web Based Mini Project	✓	✓	✓	✓	✓	✓	✓
17	Software Project Management	✓	✓	-	✓	✓	✓	✓
18	Mobile Communication	✓	✓	-	✓	✓	-	-
19	Power BI	✓	✓	✓	✓	-	✓	-

Sl. No.	Course Title	SDG4	SDG5	SDG9	SDG13
1	Network Security	✓	✓	✓	-
2	Network Security Lab	✓	✓	✓	-
3	Design and Analysis of Algorithms	✓	✓	-	-
4	Design and Analysis of Algorithms Lab	✓	✓	-	-
5	Computer Graphics & Multimedia	✓	✓	✓	-
6	Computer Graphics & Multimedia Lab	✓	✓	✓	-
7	Blockchain Technology	✓	✓	✓	-
8	Enterprise Resource Planning	✓	✓	✓	-
9	Fundamentals of Artificial Intelligence	✓	✓	✓	✓
10	AI Tools and its applications	✓	✓	✓	✓
11	Scientific Computing Techniques	✓	-	✓	-
12	Scientific Computing Techniques Lab	✓	-	✓	-
13	Big Data Analytics	✓	-	✓	✓
14	Big Data Analytics Lab	✓	-	✓	✓
15	Web Content Management System	✓	-	✓	-
16	Web Based Mini Project	✓	✓	✓	-
17	Software Project Management	✓	-	✓	-
18	Mobile Communication	✓	-	✓	-
19	Power BI	✓	-	✓	✓